ARAVALI HILLS

Land Degradation Analysis

Combined Deliverable 1 & Deliverable 2 Report

Rajasthan, India · 5 Districts · 2015–2023

Methods Applied:

Mann-Kendall Trend Test · OLS Regression · K-Means Clustering
PCA · Composite Degradation Index

Data Sources

Forest Survey of India (FSI) · Central Ground Water Board (CGWB)
Google Earth Engine — Landsat 8 NDVI & Dynamic World LULC

Project Repository & Interactive Dashboard

`https://github.com/ArvindLomrore/aravali-land-degradation`

Full source code, data pipeline, interactive HTML dashboard, and analysis notebooks

April 2026

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1 Introduction and Motivation

The Aravali Hills, stretching approximately 692 kilometres across the Indian states of Rajasthan, Haryana, Gujarat, and Delhi, constitute one of the world's oldest mountain ranges and a critical ecological barrier for northwestern India. Despite their geological antiquity, the Aravali ecosystem faces acute and accelerating anthropogenic pressure from unregulated quarrying, urban expansion, agricultural encroachment, and groundwater over-extraction.

This report presents a data-driven investigation into land degradation across five key districts situated within the Aravali region of Rajasthan: Ajmer, Bhilwara, Jaipur, Pali, and Udaipur. The study spans the period 2015 to 2023, covering a critical decade of documented ecosystem change. The overarching motivation is to quantify, characterise, and spatially differentiate degradation trajectories across this ecologically sensitive zone, providing evidence-based insights that can inform conservation policy, resource management decisions, and land rehabilitation strategies.

The Aravali range serves a dual function: as a natural windbreak preventing the eastward advance of the Thar Desert, and as a hydrological regulator sustaining groundwater recharge for millions of inhabitants across the region. Its degradation therefore carries consequences that extend well beyond ecological boundaries into food security, water availability, and climate resilience.

1.1 Objectives

- To construct a multi-source geospatial data warehouse integrating satellite, forest, and groundwater observations for the 2015–2023 period.
- To apply statistical trend analysis (Mann-Kendall), correlation and regression modelling (OLS), and unsupervised machine learning (K-Means + PCA) to identify degradation patterns across districts.
- To synthesise multi-indicator findings into a Composite Degradation Index (DI) enabling direct cross-district comparison and temporal tracking.
- To deliver an interactive analytical dashboard enabling non-technical stakeholders to explore the data dynamically.
- To discuss scalability pathways for extending this analysis to all 33 Rajasthan districts over 20+ years.

2 Problem Statement

Land degradation in the Aravali region manifests through a complex interaction of four primary mechanisms that this study tracks simultaneously:

2.1 Vegetation Loss (NDVI Decline)

The Normalised Difference Vegetation Index (NDVI), derived from Landsat 8 satellite imagery via Google Earth Engine, captures the density and health of vegetation canopy. A sustained decline in NDVI signals ecosystem stress — whether from mining-induced soil disturbance, agricultural conversion of forest fringe, or drought-exacerbated vegetation die-off. Values in this dataset range from 0.233 (Pali, 2019 — severe stress) to 0.416 (Udaipur, 2017 — healthy canopy).

2.2 Mining and Land Use Change

The Dynamic World Land Use / Land Cover (LULC) classification, processed via Google Earth Engine, identifies bare ground / mining-affected areas (Class 7) as a proxy for active quarrying and land disturbance. Rajasthan is the largest producer of dimensional stone in India, and the Aravali belt hosts extensive marble, granite, and limestone extraction operations. Bhilwara district recorded the highest single-year mining footprint at 591.9 sq km in 2015 — a value that subsequently collapsed to 59–74 sq km, suggesting either regulatory enforcement or deposit exhaustion.

2.3 Forest Cover Erosion

Forest Survey of India (FSI) biennial assessments provide district-level breakdown of Very Dense Forest (VDF), Moderately Dense Forest (MDF), Open Forest, and Scrub areas. Total forest cover is used as a structural ecosystem integrity indicator. Udaipur's 3,120 sq km 2015 baseline represents the largest forest endowment, while Bhilwara's 221–237 sq km range reflects chronic degradation from decades of industrial activity. Forest change percentages reveal that Jaipur lost 14.15% of its forest cover between 2015 and 2023.

2.4 Groundwater Depletion

Central Ground Water Board (CGWB) monitoring data provides annual average groundwater depth measurements. Increasing depth-to-water-table indicates aquifer depletion — a direct consequence of over-extraction driven by agricultural demand, urban consumption, and disrupted recharge from forest loss. Jaipur and Pali both record consistently deep groundwater (30–35 m) across all five observation years — a structural stress condition that correlates with their classification as high-degradation districts.

3 Data Sources and Collection

3.1 Satellite Imagery — Google Earth Engine

Landsat 8 Surface Reflectance composites were processed within Google Earth Engine (GEE) to generate biennial mean NDVI rasters for 2015, 2017, 2019, 2021, and 2023. The NDVI formula (NIR − Red)/(NIR + Red) using Bands 5 and 4 respectively was applied over cloud-masked composites for each monsoon season (June–September), when vegetation response is maximal and cloud contamination is most challenging.

Dynamic World probabilistic LULC classification was used to extract bare ground / potential mining areas (Class 7) per district per year using zonal statistics via the `rasterstats` Python library. All rasters were processed at 30 m spatial resolution.

3.2 Forest Survey of India (FSI)

FSI biennial State of Forest Reports provide tabular district-level forest cover statistics in square kilometres. Data was compiled for five observation years corresponding to satellite data availability. Forest categories recorded include Very Dense Forest (VDF, canopy density > 70%), Moderately Dense Forest (MDF, 40–70%), Open Forest (10–40%), and Scrub (< 10% canopy).

3.3 Central Ground Water Board (CGWB)

CGWB district-level groundwater depth monitoring data was obtained for all nine years between 2015 and 2023. Only odd-year values corresponding to satellite data were retained for multi-indicator analysis. The Ajmer 2023 value of 208.40 m was identified as a data anomaly (80.8σ above district historical mean) and imputed using the district median (9.11 m) with a transparency flag (`GW_Imputed = True`).

3.4 Dataset Summary

Table 1: Summary of data sources, variables, and formats

Source	Variables	Temporal Coverage	Spatial Unit	Format
GEE (Landsat 8)	Mean NDVI	2015, 17, 19, 21, 23	District	GeoTIFF → CSV
GEE (Dynamic World)	Mining/Bare Area (sq km)	2015, 17, 19, 21, 23	District	GeoTIFF → CSV
Forest Survey of India	VDF, MDF, Open Forest, Scrub, Total Forest	2015, 17, 19, 21, 23	District	CSV
Central Ground Water Board	GW Depth (m)	2015–2023 (annual)	District	CSV

4 Data Pipeline and Warehouse Design

4.1 ETL Pipeline

The data engineering pipeline was implemented in Python using GeoPandas, Rasterio, and Rasterstats for spatial processing, and Pandas + SQLite for tabular storage. The pipeline proceeds in three stages:

- **Extract:** GEE rasters exported as GeoTIFFs; district boundary shapefiles (ADM2 level) obtained from GADM.
- **Transform:** Zonal statistics computed per district per year; forest and groundwater CSVs cleaned and standardised; anomaly detection and imputation applied; feature engineering executed.
- **Load:** Consolidated fact table (`aravali_master_fact_table.csv`, 45 rows × 10 columns) loaded into a SQLite data warehouse with a star schema.

4.2 Star Schema Design

Table 2: Data warehouse star schema

Table	Type	Key Columns	Purpose
Dim_Time	Dimension	Time_ID, Year, Era	Temporal dimension (Pre/Post-2020)
Dim_Location	Dimension	Location_ID, District_Name, State	Spatial dimension (5 districts)
Fact_Degradation	Fact	Fact_ID, Time_ID, Location_ID	Core metrics: NDVI, GW, Mining, Forest
Fact_Land_Distribut	Fact	Distribution_ID, Location_ID, Time_ID	LULC class areas per district-year
Dim_Raster_Registry	Dimension	Raster_ID, Year, Layer_Type	Metadata for 10 source rasters

4.3 Phase 1 Data Cleaning

The raw master fact table contained 45 rows spanning all nine years (2015–2023). Key cleaning steps: even years were excluded for ML analysis, reducing to 25 complete observations (5 × 5). The Ajmer 2023 GW anomaly ($z = 80.8\sigma$) was imputed to the district median (9.11 m) with a `GW_Imputed` flag. Five derived features were engineered: `NDVI_Anomaly`, `Forest_Density_Score`, `Forest_Change_pct`, `Mining_Forest_Ratio`, and `GW_Stress_Flag` (binary: depth > 30 m).

5 Exploratory Data Analysis (Deliverable 1)

5.1 District-Level Overview

Initial exploratory analysis revealed significant heterogeneity across the five study districts. Key summary statistics for the 2015–2023 period are shown in Table 3.

Table 3: District-level summary statistics, 2015–2023

District	Mean NDVI	Mean Mining (sq km)	Mean Forest (sq km)	Mean GW (m)	GW Stress Yrs
Ajmer	0.275	117.7	310.5	8.6	0/5
Bhilwara	0.313	187.9	226.7	8.3	0/5
Jaipur	0.277	230.0	573.4	32.8	5/5
Pali	0.283	124.1	686.7	32.8	5/5
Udaipur	0.371	38.3	2,832.2	6.6	0/5

5.2 Time-Series Trends

Figure 1 shows time-series trends across all four key metrics for the five Aravali districts from 2015 to 2023. Bold solid lines indicate statistically significant Mann-Kendall trends. Jaipur's 2021 mining surge (438.2 sq km) and Bhilwara's anomalous 2015 mining peak (591.9 sq km) are clearly visible. Udaipur consistently records the highest NDVI and lowest mining activity throughout the study period.

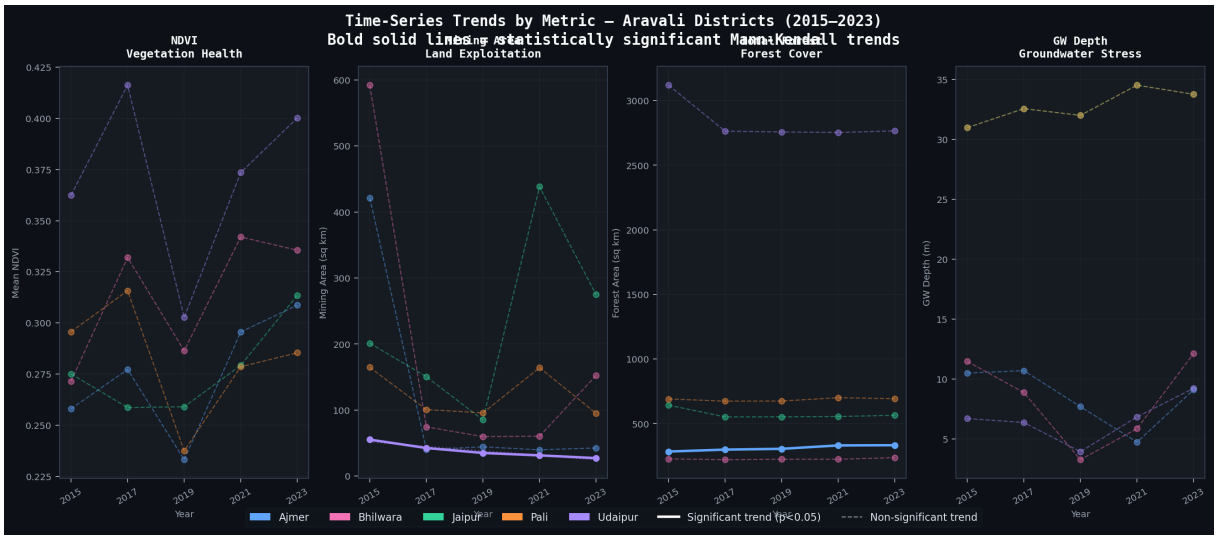

Figure 1: Time-series trends for NDVI, Mining Area, Forest Cover, and Groundwater Depth across all five Aravali districts (2015–2023). Bold solid lines denote statistically significant Mann-Kendall trends ($p < 0.05$); dashed lines indicate non-significant trends.

6 Advanced Analysis (Deliverable 2)

6.1 Mann-Kendall Trend Analysis

The Mann-Kendall test is a non-parametric rank-based method for detecting monotonic trends in time-series data, particularly suited for environmental datasets with non-normal

distributions and potential outliers. It was implemented from scratch using the standard formulation:

$$S = \sum_{j>k} \text{sign}(x_j - x_k) \tag{1}$$

$$\tau = \frac{S}{n(n-1)/2}, \qquad \text{Var}(S) = \frac{n(n-1)(2n+5)}{18} \tag{2}$$

P-values were computed using a manual approximation of the standard normal CDF (Abramowitz & Stegun rational approximation). With $n = 5$ time points per series, the critical value $|S| = 10$ ($\tau = \pm 1.0$) is required for significance at $\alpha = 0.05$.

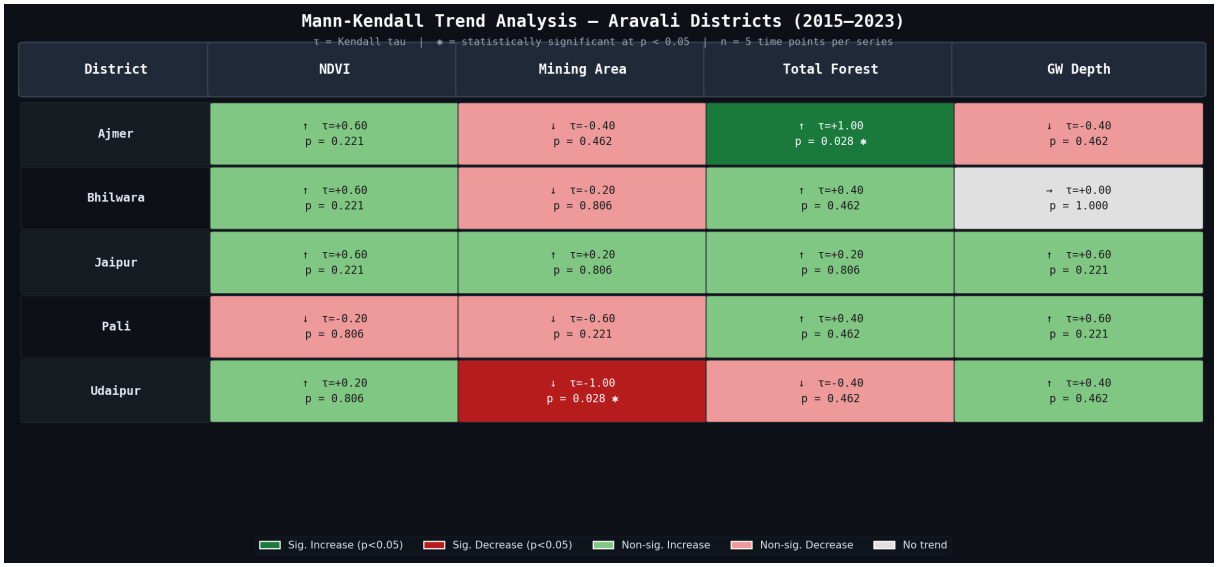

Figure 2: Mann-Kendall trend summary table — trend direction, Kendall τ, and p-value for all 20 district-metric combinations. Cells highlighted in dark green (significant increase) or dark red (significant decrease) are statistically significant at $p < 0.05$.

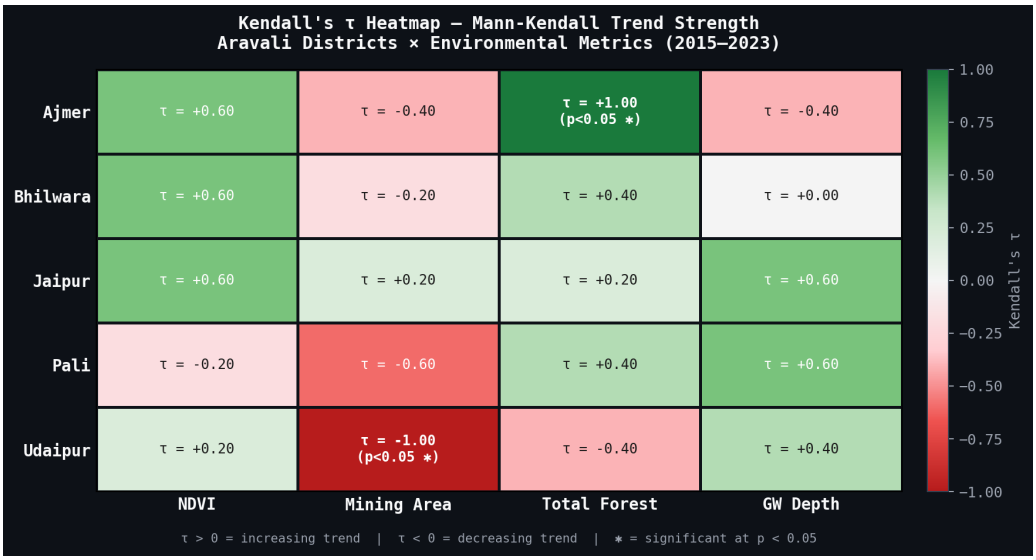

Figure 3: Kendall's τ heatmap showing trend strength and direction across all district-metric combinations. Cells annotated with $*$ are statistically significant at $p < 0.05$. Green = increasing trend; red = decreasing trend; intensity reflects magnitude of τ.

6.1.1 Key Findings

Table 4: Selected Mann-Kendall trend analysis results

District	Metric	Kendall τ	p-value	Interpretation
Ajmer	Total Forest	+1.000	0.027	**Significant:** Perfect monotonic forest growth (+17.3% by 2023)
Udaipur	Mining Area	−1.000	0.027	**Significant:** Perfect monotonic mining contraction (−50.6%)
Jaipur	GW Depth	+0.600	0.221	Non-sig: Strong increasing trend — deepening aquifer stress
Pali	Mining Area	−0.600	0.221	Non-sig: Consistent mining reduction
Bhilwara	GW Depth	0.000	1.000	No trend: GW fluctuates without directional pattern

The limited statistical power of Mann-Kendall with $n = 5$ is an acknowledged limitation. Significance is only achievable when the trend is perfectly monotonic (all 10 pairwise

comparisons concordant). Non-significant results do not imply absence of trend — the ecological interpretation of $\tau = +0.6$ for Jaipur and Pali GW depth remains substantively meaningful.

6.2 Pearson Correlation and OLS Regression

OLS regression was implemented from scratch using the closed-form solution for simple linear regression. P-values were computed using a manual t-distribution CDF approximation via continued fraction expansion.

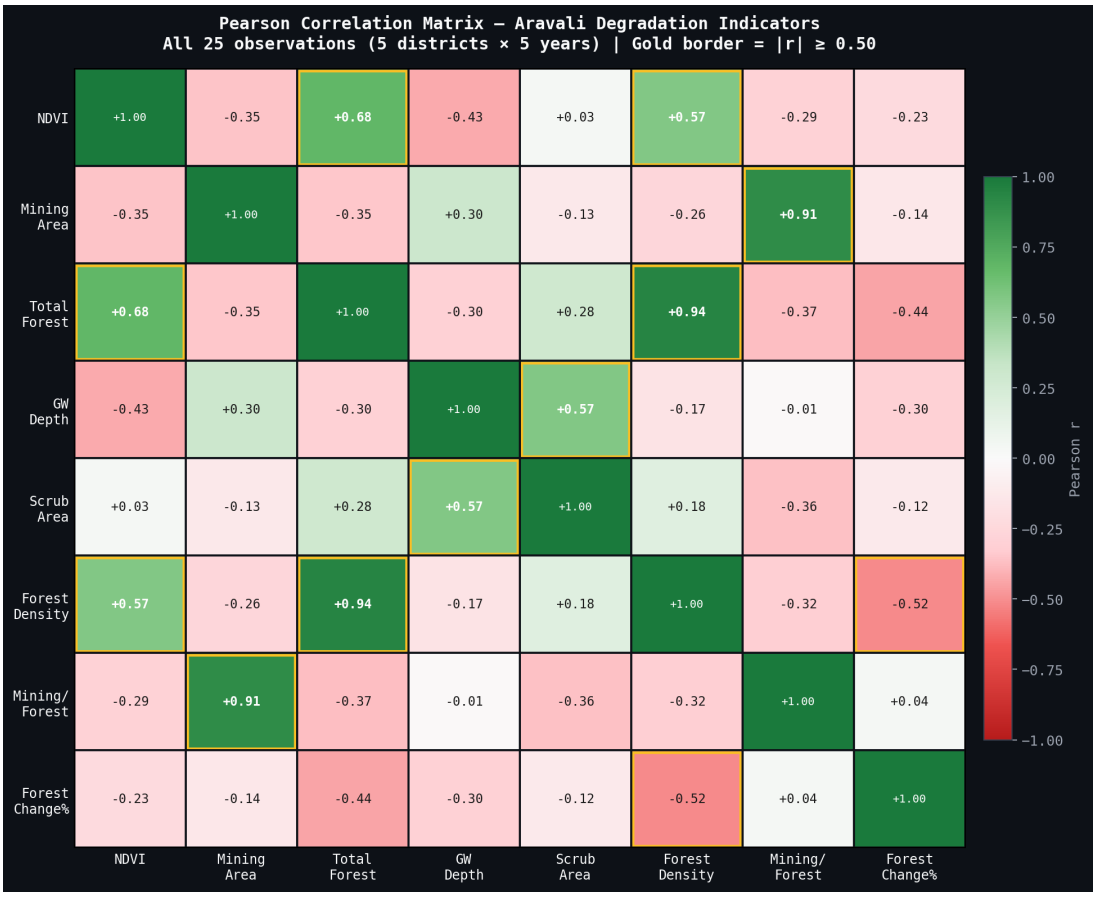

Figure 4: Pearson correlation matrix across eight environmental indicators for all 25 observations (5 districts × 5 years). Gold borders highlight correlations with $|r| \geq 0.50$. The Total Forest ↔ Forest Density correlation ($r = +0.94$) and Total Forest → NDVI ($r = +0.68$) are the strongest relationships in the dataset.

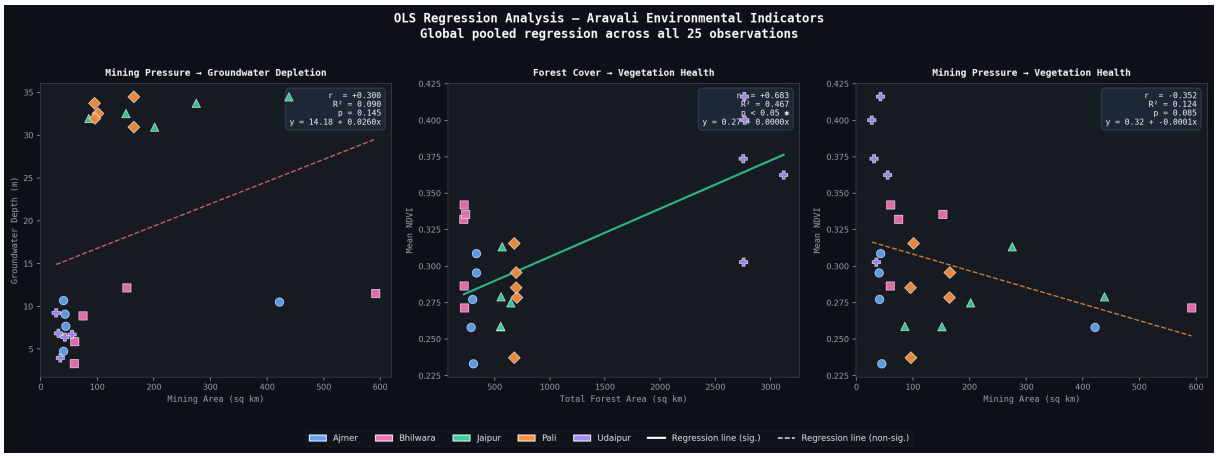

Figure 5: OLS regression panels — Mining Area → GW Depth (left), Forest Cover → NDVI (centre), Mining Area → NDVI (right). Solid regression lines indicate statistical significance ($p < 0.05$); dashed lines are non-significant. District colour-coding reveals that the Udaipur outlier effect partly drives pooled correlations.

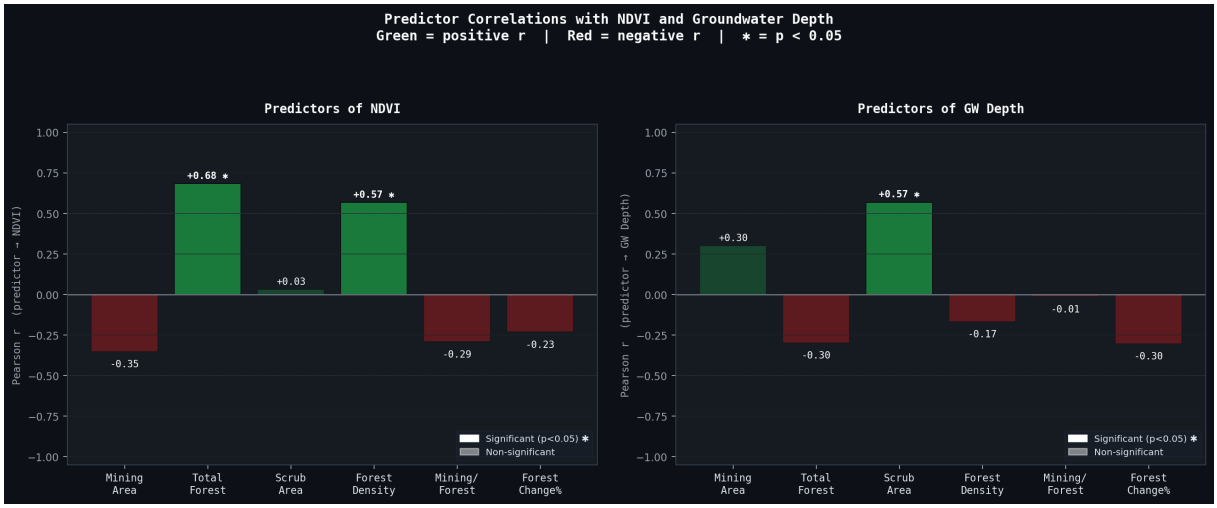

Figure 6: Predictor correlations with NDVI (left) and Groundwater Depth (right). Green bars indicate positive Pearson r; red bars indicate negative r. Bars marked $*$ are significant at $p < 0.05$. Total Forest ($r = +0.68$) and Forest Density ($r = +0.57$) are the strongest predictors of NDVI.

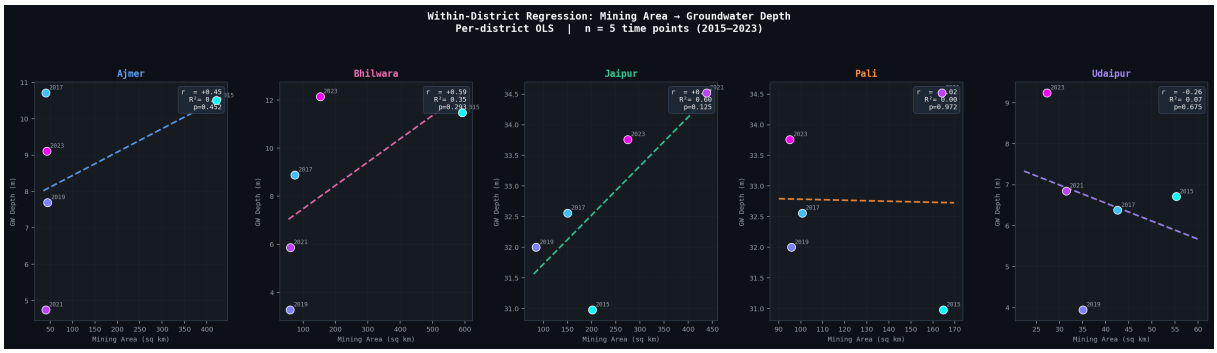

Figure 7: Within-district OLS regression for Mining Area → Groundwater Depth ($n = 5$ per district). Year labels provide temporal context. Jaipur shows the strongest single-district mining-groundwater relationship ($r = +0.77$), consistent with the hypothesis that mining activity amplifies aquifer stress in high-pressure urban districts.

6.2.1 Key Correlations

Table 5: Key Pearson correlations and OLS regression results

Variable Pair	Pearson r	R^2	p-value	Significance
Total Forest → NDVI	+0.683	0.467	0.0002	**Significant**
Mining Area → NDVI	−0.352	0.124	0.085	Non-significant
Mining Area → GW Depth	+0.300	0.090	0.145	Non-significant
Forest Density → NDVI	+0.568	0.323	< 0.01	**Significant**
Total Forest ↔ Forest Density	+0.943	0.889	< 0.001	**Very strong**

The Forest → NDVI relationship explains 46.7% of NDVI variance globally ($R^2 = 0.467$, $p = 0.0002$), confirming that structural forest cover is the primary predictor of ecosystem health. The weak pooled Mining → GW correlation ($r = +0.300$) reflects structural confounding by urban extraction pressure in Jaipur and Pali.

6.3 K-Means Clustering and Manual PCA

K-Means clustering was applied to a district-level feature matrix comprising eight change-rate and level indicators. Both K-Means and PCA were implemented from scratch — K-Means used 100 random restarts with K-Means++ initialisation; PCA used power iteration eigendecomposition with deflation.

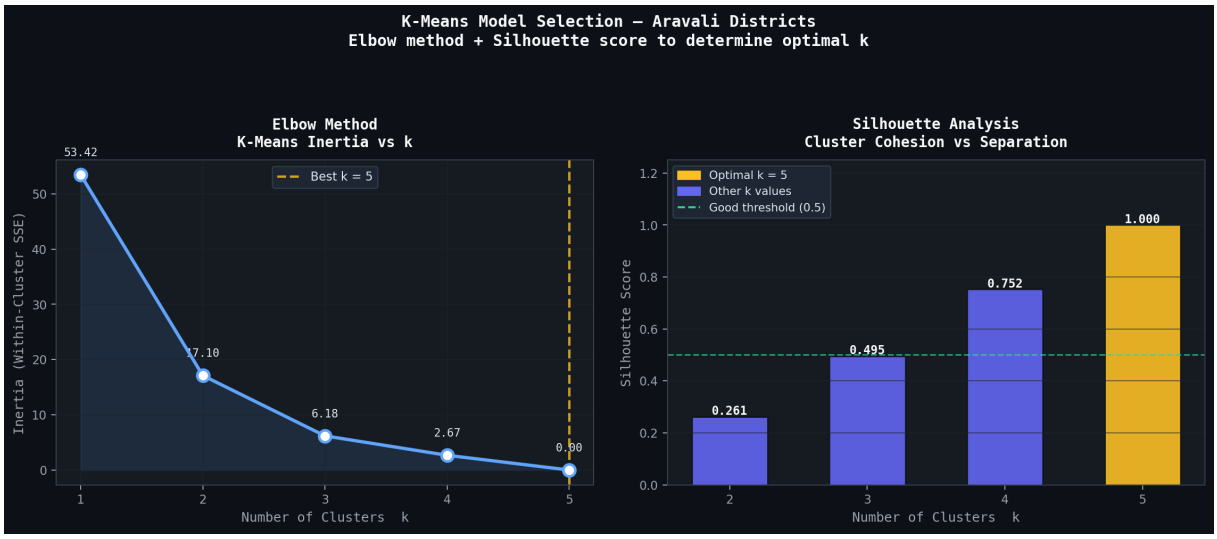

Figure 8: K-Means model selection — Elbow method (left) and Silhouette score analysis (right) for $k = 1$ to $k = 5$. Although silhouette scores peak at $k = 5$, this produces singleton clusters. $k = 3$ was selected as the optimal ecological partition based on interpretability and cluster cohesion (silhouette = 0.495).

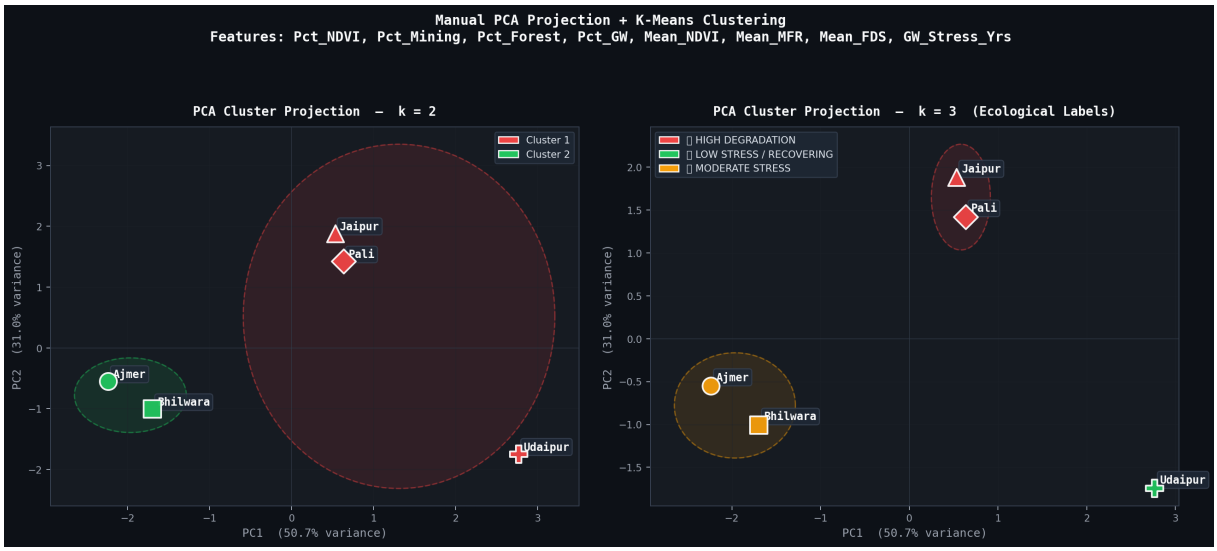

Figure 9: Manual PCA projection of five districts onto two principal components (81.7% combined variance). PC1 (50.7% variance) separates districts by forest scale and NDVI level; PC2 (31.0% variance) captures change dynamics and GW stress. $k = 3$ cluster boundaries (right) confirm ecologically meaningful groupings with clean separation in PCA space.

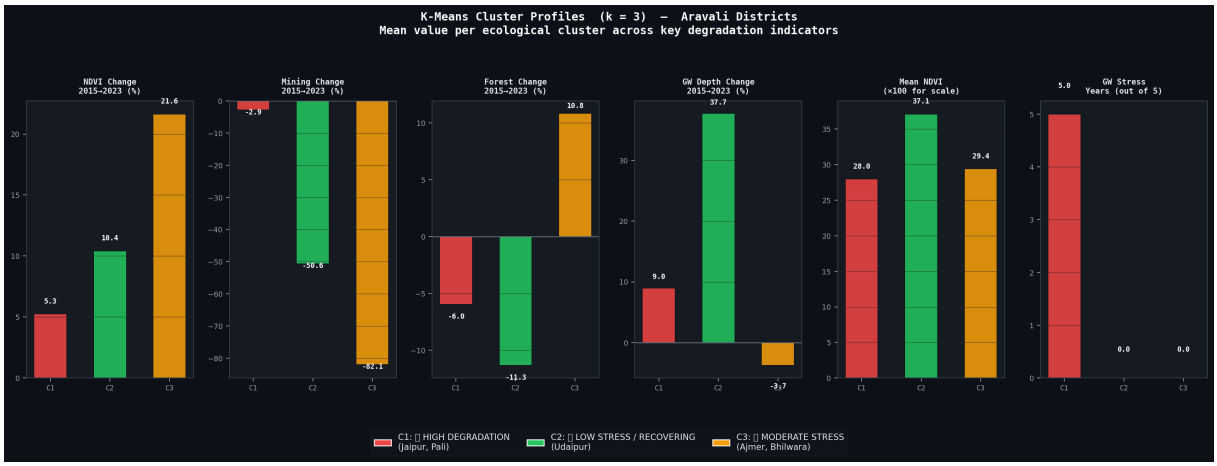

Figure 10: K-Means cluster profiles ($k = 3$) — mean value per ecological cluster across six key degradation indicators. C1 (HIGH DEGRADATION: Jaipur, Pali) shows elevated GW stress years and modest NDVI change; C2 (LOW STRESS: Udaipur) shows the greatest mining contraction (−50.6%); C3 (MODERATE STRESS: Ajmer, Bhilwara) shows the strongest NDVI recovery (+21.6%).

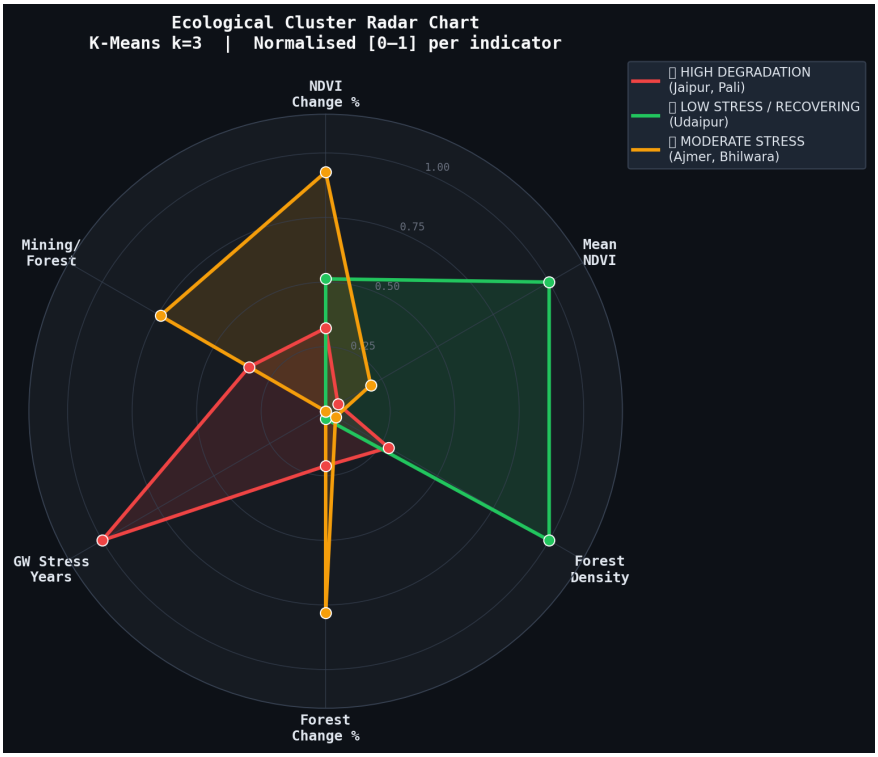

Figure 11: Ecological Cluster Radar Chart — normalised $[0, 1]$ profiles across six indicators for $k = 3$ clusters. The HIGH DEGRADATION cluster (red) dominates in GW Stress Years; the LOW STRESS cluster (green) dominates in Mean NDVI and Forest Density; the MODERATE STRESS cluster (orange) shows the highest Mining/Forest ratio, reflecting Bhilwara's historical industrial activity.

6.3.1 Cluster Assignments ($k = 3$)

Table 6: K-Means cluster assignments and ecological labels

Cluster	Ecological Label	Districts	Defining Characteristics
1	HIGH DEGRADATION	Jaipur, Pali	GW Stress: 5/5 years; Mean GW: 32.8 m; Forest loss
2	LOW STRESS / RECOVERING	Udaipur	Mean NDVI: 0.371; Mining declining ($\tau = -1.0$); 3,120 sq km forest baseline
3	MODERATE STRESS	Ajmer, Bhilwara	GW Stress: 0/5; Forest recovering in Ajmer (+17%); High historical mining in Bhilwara

6.4 Composite Degradation Index

The Composite Degradation Index (DI) synthesises all four environmental pillars into a single scalar score per district per year, aligned with the UNCCD Land Degradation Neutrality (LDN) framework:

$$\text{DI} = 0.30 \times \text{DI_NDVI} + 0.25 \times \text{DI_Mining} + 0.25 \times \text{DI_Forest} + 0.20 \times \text{DI_GW} \quad (3)$$

All components are normalised to $[0, 1]$ via global min-max scaling across all 25 observations.

Table 7: DI component construction and weights

Component	Raw Metric	Normalisation	Weight	Rationale
DI_NDVI	Mean NDVI	1 − min-max(NDVI)	30%	Primary vegetation health signal
DI_Mining	Mining Area (sq km)	min-max(Mining)	25%	Active anthropogenic disturbance
DI_Forest	Total Forest (sq km)	1 − min-max(Forest)	25%	Structural ecosystem component
DI_GW	GW Depth (m)	min-max(GW Dep	20%	Lagged subsurface consequence

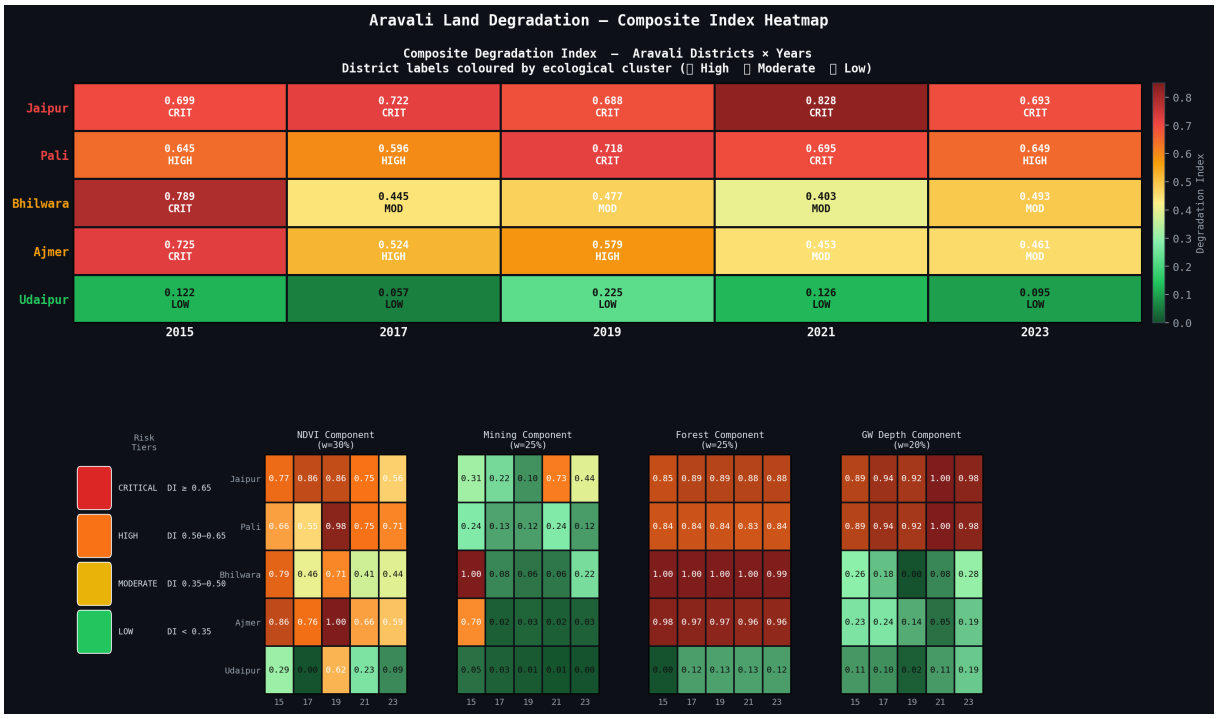

Figure 12: Composite Degradation Index heatmap (top) and component breakdown sub-heatmaps (bottom) for all five districts across 2015–2023. Jaipur's persistent CRITICAL status and Udaipur's stable LOW status are the most striking structural contrasts. Bhilwara's transition from CRITICAL (2015) to MODERATE (2017 onward) is clearly visible following the collapse of its 2015 mining anomaly.

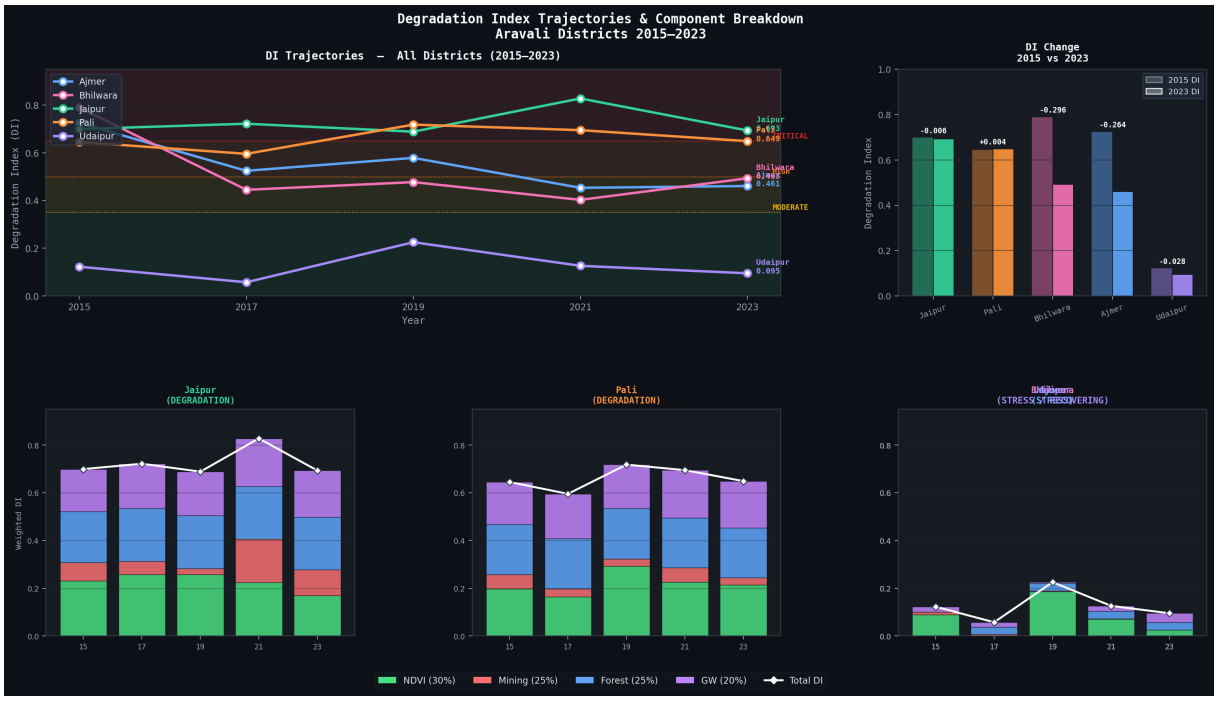

Figure 13: Degradation Index trajectories for all five districts (2015–2023) with DI change comparison (2015 vs. 2023). Risk threshold lines at DI = 0.35 (MODERATE), 0.50 (HIGH), and 0.65 (CRITICAL) are shown. Stacked component breakdown panels (lower row) show the weighted contribution of NDVI, Mining, Forest, and GW components to each district's total DI per year.

6.4.1 Risk Tier Classification

Table 8: DI risk tier classification across 25 observations

Tier	DI Range	Colour	Observations	Interpretation
CRITICAL	≥ 0.65	Red	9/25 (36%)	Severe multi-indicator stress; urgent intervention required
HIGH	0.50 – 0.65	Orange	5/25 (20%)	Multiple degradation pressures; monitoring essential
MODERATE	0.35 – 0.50	Yellow	6/25 (24%)	Active degradation; preventive management recommended
LOW	< 0.35	Green	5/25 (20%)	Ecosystem relatively intact; sustain conservation measures

Aravali Districts — Degradation Risk Ranking (2023 Snapshot)

Rank	District	DI 2023	Risk Tier	NDVI Comp	Mining Comp	Forest Comp	GW Comp	ΔDI 15–23	Trend	Ecological Cluster
1	Jaipur	0.6934	CRITICAL	0.561	0.438	0.881	0.976	-0.0059	→ Stable	HIGH DEGRADATION
2	Pali	0.6488	HIGH	0.714	0.120	0.837	0.976	+0.0037	→ Stable	HIGH DEGRADATION
3	Bhilwara	0.4930	MODERATE	0.441	0.222	0.994	0.284	-0.2963	▼ Improving	MODERATE STRESS
4	Ajmer	0.4607	MODERATE	0.588	0.027	0.961	0.187	-0.2638	▼ Improving	MODERATE STRESS
5	Udaipur	0.0949	LOW	0.088	0.000	0.122	0.191	-0.0276	▼ Improving	LOW STRESS / RECOVERING

DI components weighted: NDVI 30% · Mining 25% · Forest 25% · GW 20% | Normalised globally across all 25 observations | ★ Ajmer 2023 GW imputed (see Phase 1 audit)

Figure 14: 2023 Degradation Risk Ranking — all five Aravali districts ranked by DI score, with risk tier, individual component values, ΔDI trend (2015–2023), and ecological cluster assignment. Jaipur's GW component score of 0.976 is the highest in the dataset, anchoring its CRITICAL classification despite no worsening mining trend.

7 Scalability, Memory, and Performance

7.1 Current System Profile

The current implementation operates on 5 districts $\times$ 5 biennial time points $=$ 25 complete observations, processing in-memory in under 30 seconds. The SQLite warehouse occupies less than 1 MB on disk.

Table 9: Current system scalability profile

Dimension	Current Scale	Bottleneck?	Notes
Districts	5	No	SQLite scales to thousands without modification
Years	5 (biennial)	No	Annual data would add 9 more rows per district
Raster Files	10	Yes at scale	Rasterstats is single-threaded; 33 districts would require 20–30 min
Observations (ML)	25 rows	Yes for DL	K-Means and OLS work well; insufficient for neural networks
Storage	$<$ 1 MB	No	Raster TIFFs add 50–200 MB per image at Rajasthan scale

7.2 Identified Limitations

- $n = 5$ per time series limits Mann-Kendall power; annual data ($n = 9$) would lower the significance threshold to $|S| = 14$.
- 30 m Landsat pixels may not capture fine-scale mining pit boundaries; Sentinel-2 (10 m) would improve accuracy but increase data volume 9×.
- Jaipur and Pali appear to share the same CGWB monitoring station, yielding identical GW values — confounding per-district groundwater analysis.
- The Ajmer 2023 GW anomaly (208.40 m) required imputation; the root cause (likely a unit error) could not be verified without raw station records.
- K-Means on $n = 5$ points is inherently unstable; cluster boundaries should be treated as exploratory rather than definitive.

7.3 Future Scalability Pathway

Scale-up should proceed along three tracks. For the *data pipeline*, GEE server-side zonal statistics should replace local Rasterstats processing, SQLite should migrate to PostgreSQL + PostGIS, and Cloud-Optimised GeoTIFF (COG) format on AWS S3/GCS should be adopted for raster storage. For *analysis*, expanding to 33 districts × 20 years (660 observations) enables Random Forest regression, DBSCAN clustering, STL time-

series decomposition, and Moran's I spatial autocorrelation. For *deployment*, the self-contained HTML dashboard can be hosted on Hugging Face Spaces at zero infrastructure cost, and a Gradio wrapper would enable non-technical users to submit their own district data for automated assessment.

8 Interactive Analysis Dashboard

An interactive web-based dashboard was developed as the primary user interface, replacing a server-dependent Gradio application with a fully self-contained HTML file deployable in any modern browser without installation or server infrastructure.

8.1 Technical Architecture

- **Technology:** Single-file HTML + CSS + JavaScript using Chart.js (CDN-loaded).
- **Data:** All 25 observations, MK results, cluster assignments, and DI values embedded directly in the JavaScript data layer.
- **Analytics:** Pearson correlation and OLS regression computed client-side in real time.
- **Access:** Available via the project GitHub repository below.

Project Repository & Interactive Dashboard

`https://github.com/ArvindLomrore/aravali-land-degradation`

The repository contains the complete ETL pipeline, SQLite data warehouse, all analysis scripts (Mann-Kendall, OLS, K-Means, PCA, DI), the self-contained interactive HTML dashboard, and reproducible Jupyter notebooks for all figures in this report.

8.2 Dashboard Structure — Three Tabs

Tab 1: District Explorer — Dropdown selectors for district and metric; four dynamic stat cards (2023 value, 2015→2023 delta, DI score, significant MK trend count); Mann-Kendall trend cards; full interactive multi-district time-series chart; four mini metric panels per district.

Tab 2: Correlation Explorer — Any pairwise X/Y metric combination via dropdowns; district filter; live scatter plot with per-district colour/marker coding and OLS regression line (solid = significant, dashed = non-significant); live β_0, β_1, r, R^2 panel; bar chart of all predictor correlations vs NDVI; 6×6 correlation heat grid.

Tab 3: Degradation Dashboard — Ecological cluster summary cards; year selector; interactive DI heatmap with hover; DI trajectory chart with risk threshold annotations; stacked component breakdown bar chart; full sortable risk ranking table with tier badges and cluster assignments.

9 Key Findings and Conclusions

9.1 Summary of Findings

This study applied a multi-method analytical framework to characterise land degradation across five Aravali Hill districts in Rajasthan over the 2015–2023 period. The principal findings are:

- **Jaipur** is the most critically degraded district (DI = 0.693 in 2023, CRITICAL tier), driven primarily by structural groundwater depletion (30–35 m depth across all years) with no improving trajectory.

- **Pali** shares Jaipur's groundwater stress profile and high-degradation cluster membership, suggesting shared aquifer dynamics and structural risk despite lower mining activity.

- **Bhilwara** exhibited the sharpest improvement — from CRITICAL in 2015 (DI = 0.789) to MODERATE by 2017 — demonstrating that acute mining disturbance can contract rapidly under regulatory or market pressure.

- **Ajmer** shows statistically significant forest growth (MK $\tau = +1.0$, $p = 0.027$), the only confirmed ecological improvement. Forest area grew 17.3% from 284 sq km (2015) to 333 sq km (2023).

- **Udaipur** is the ecological anchor of the region (DI range: 0.057–0.225, LOW tier), anchored by 3,120 sq km of forest and the only confirmed positive trend: monotonic mining decline (MK $\tau = -1.0$, $p = 0.027$).

- **Forest cover** is the strongest predictor of ecosystem health: Total Forest $\rightarrow$ NDVI yields $r = +0.683$, $R^2 = 0.467$, $p = 0.0002$ — the only globally significant regression in the study.

- **K-Means** ($k = 3$) successfully partitions the five districts into ecologically meaningful groups, confirmed by PCA (81.7% combined variance explained in 2D).

9.2 Policy Implications

- **Groundwater management in Jaipur and Pali** requires immediate intervention. The 30+ m depth-to-water-table cannot be reversed through forest conservation alone; dedicated aquifer recharge programmes and urban extraction regulation are essential.

- **Bhilwara's recovery** demonstrates the effectiveness of mining regulation. Sustaining the post-2015 reduction should be prioritised, with enforcement mechanisms to prevent reversion.

- **Ajmer's forest recovery** (+17.3%) should be documented, attributed mechanistically, and replicated across the region.

- **Udaipur's forest reserve** should be formally protected. Any mining expansion in Udaipur would disproportionately threaten the region's only ecological resilience anchor.

9.3 Limitations and Future Work

The principal limitations are the small temporal sample ($n = 5$ per district), shared groundwater monitoring data for Jaipur and Pali, the imputed Ajmer 2023 GW value, and the absence of socioeconomic variables that would enable causal rather than correlational analysis. Future work should: (a) extend temporal coverage to annual observations; (b) expand to all 33 Rajasthan districts; (c) incorporate Sentinel-2 10 m imagery; and (d) develop a predictive degradation model using Random Forest regression on the expanded dataset.

Aravali Hills Land Degradation Analysis · Rajasthan, India · 2015–2023

`https://github.com/ArvindLomrore/aravali-land-degradation`